

Article

Framework for the Evaluation of Nap-Compatible Classroom Chairs

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Abstract

Recent policy initiatives, such as the Ministry of Education of China's guidelines on student sleep management, together with international empirical studies on daytime rest, highlight that appropriate opportunities for midday rest can support students' attention, memory, and overall learning. Motivated by these developments, this study introduces a conflict-sensitive, multidimensional framework for evaluating nap-compatible classroom chairs. The proposed PLEASURC framework integrates eight evaluation dimensions, organized into four domains—ergonomic, operational, perceptual, and affective—thereby combining both expert and student perspectives across three use scenarios (writing, adjusting, and napping). Perceptual divergence between groups is addressed through Jensen–Shannon divergence-based adaptive weighting, which adjusts expert–student influence according to their agreement level. Scenario-sensitive patterns are extracted via tensor decomposition (a statistical factor analysis technique), while SHAP-enhanced machine learning (an explainable model interpretation method) is employed to identify the most influential predictors of perceived comfort. Findings indicate that relational and emotional dimensions (e.g., Relatability, Learnability, Aesthetics) significantly influence perceived comfort, surpassing structural considerations alone. The study also demonstrates a closed feedback loop from evaluation to redesign, supporting the practical utility of the framework through the optimized chair prototype. Overall, this research offers a replicable and interpretable framework for ergonomic evaluation and data-driven redesign of multifunctional school furniture, contributing to both student-centered learning environments and sustainable educational infrastructure.



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Keywords: classroom furniture; ergonomic evaluation; perceptual divergence; Jensen–Shannon divergence; student–expert feedback; SHAP; tensor decomposition; human-centered design; nap-compatible design

1. Introduction

Indoor learning environments substantially influence children's cognitive performance, emotional well-being, and physical health. In certain education systems, such as China, midday rest has been formally recognized and safeguarded through national and municipal guidelines. The Ministry of Education issued official sleep-management regulations in 2021 [1], requiring schools to allocate sufficient sleep opportunities, including a brief lunch break where conditions allow. Local governments, such as Beijing and Shenzhen, have subsequently reinforced these requirements through inspection standards and design guidelines. Within these contexts, evaluating nap-compatible classroom furniture is not a universal prescription but rather a policy-enabled practice relevant to specific jurisdictions. International sleep research also supports this rationale: short daytime rest has been

associated with improved attention regulation, memory consolidation, and overall learning in children and adolescents [2–4]. Thus, where midday rest is institutionally scheduled, classroom furniture ergonomics become an important factor for both academic engagement and restorative recovery.

Despite policy recognition and scientific evidence, conventional classroom furniture rarely incorporates ergonomic or affective design considerations for rest-related activities. Existing systems are typically optimized for upright learning tasks, neglecting the dual requirement of supporting both study and rest. This design gap may hinder not only comfort but also the effective implementation of sustainable school health practices [5,6].

Existing ergonomic evaluation frameworks have been largely developed for adult-oriented, task-specific environments, prioritizing productivity and biomechanical efficiency. Such models often overlook the multisensory experiences and emotional needs unique to children [7,8]. This mismatch is particularly evident in educational contexts, where furniture should support not only academic engagement but also restorative rest, demanding adaptability to children's posture, ease of use, and psychological comfort. Therefore, there is an urgent need for a child-centered evaluation framework that systematically integrates both structural performance and affective experiences to address this gap in current classroom design.

To address this gap, we introduce the PLEASURC model—a behaviorally grounded, multidimensional framework encompassing structural, operational, perceptual, and affective domains. The model's eight dimensions—Ergonomics (E), Safety (S), Learnability (L), Usability (U), Perception (P), Aesthetics (A), Relatability (R), and Comfort (C)—are systematically derived from interdisciplinary research and international standards, including ISO 9241-210 on human–system interaction, Kansei engineering, emotional design theory, and validated ergonomic assessment frameworks. Distinct from conventional methods that apply static evaluation criteria, the PLEASURC framework incorporates a conflict-sensitive weighting mechanism based on Jensen–Shannon divergence to dynamically reconcile expert and student perspectives. Furthermore, tensor decomposition is employed to reveal scenario-specific sensitivity patterns, and SHAP-enhanced machine learning is used to interpret the key drivers of perceived comfort. Together, these methods provide a comprehensive and interpretable approach to assessing nap-compatible classroom chairs in student-centered learning environments [9].

By integrating conflict-sensitive weighting, scenario-driven factor modeling, and explainable machine learning, this study develops a rigorous paradigm for evaluating multifunctional school furniture for children. Addressing the gap between policy-supported midday rest and the lack of ergonomic or emotional accommodation in classroom furniture, the PLEASURC framework provides both methodological rigor and practical design guidance, offering a replicable approach to align child-centered ergonomics with sustainable educational infrastructure.

2. Materials and Methods

2.1. Development of the PLEASURC Evaluation Framework

To systematically evaluate nap-compatible classroom chairs from a user-centered and ergonomically grounded perspective, we developed the PLEASURC framework—a structured, multidimensional model specifically designed for children's rest-related activities in indoor educational settings [10]. The framework comprises eight evaluation dimensions organized into the following four conceptual domains:

- Structural domain: Ergonomics (E), Safety (S);
- Operational domain: Learnability (L), Usability (U);
- Perceptual domain: Perception (P), Aesthetics (A);

- Affective domain: Relatability (R), Comfort (C).

The conceptual basis was derived from a synthesis of the following:

1. International ergonomic standards (e.g., ISO 9241-210 on human–system interaction; GB/T 28007-2011 on children’s furniture) [10–12];
2. Emotional and perceptual modeling theories, including Kansei Engineering and Norman’s Emotional Design theory;
3. Empirical studies on child-centered ergonomics and affect-sensitive design in educational contexts.

These four domains can also be described as relational dimensions, since each represents a distinct relationship between the child and the furniture: the body–product relationship (structural), the action–product relationship (operational), the sense–product relationship (perceptual), and the emotion–product relationship (affective). Each dimension addresses a distinct yet complementary aspect of user experience. Ergonomics and Safety focus on biomechanical support and injury prevention; Learnability and Usability emphasize intuitive operation and efficiency; Perception and Aesthetics address sensory appeal and visual-tactile harmony; Relatability and Comfort capture psychosomatic ease and emotional bonding—often overlooked in traditional assessments but vital for child well-being. This relational organization clarifies how physical support, functional operation, sensory appraisal, and emotional acceptance interact to shape overall comfort.

This taxonomy reflects the dual functional nature of modern classroom furniture, which should simultaneously support learning and rest. Accordingly, the PLEASURC model provides a theoretically coherent and contextually relevant framework for evaluating multifunctional school furniture in research-integrated learning environments [13,14]. Definitions and classifications of the dimensions are summarized in Table 1, while Figure 1 visualizes the four-domain structure.

Table 1. Definitions of the PLEASURC evaluation dimensions and their classification by domain.

Domain	Dimension	Definition
Structural	Ergonomics (E)	The extent to which the desk–chair supports proper posture, spinal alignment, and reduces physical strain.
	Safety (S)	The perceived structural stability, material safety, and protection from hazards such as pinching, tipping, or sharp edges.
Operational	Learnability (L)	The ease with which children can understand and initiate functional operations without external instruction.
	Usability (U)	The intuitiveness, efficiency, and fault-tolerance of the desk–chair’s features in classroom use.
Perceptual	Perception (P)	The tactile and thermal sensations of surface materials that influence comfort.
	Aesthetics (A)	The visual qualities (shape, color, style) that affect appeal and perceived quality.
Affective	Relatability (R)	The emotional connection users feel, based on familiarity, friendliness, or playful design.
	Comfort (C)	The overall physical and psychological ease experienced during use.

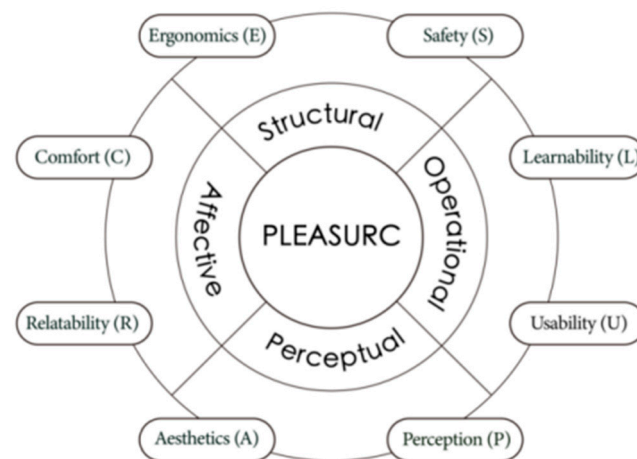


Figure 1. PLEASURC: A Four-Layer Evaluation Model.

2.2. Experimental Design and Scenario Configuration

A controlled experiment was designed to evaluate nap-compatible classroom chairs under realistic usage conditions, integrating scenario-based tasks, randomized product exposure, and structured feedback collection from both students and experts.

2.2.1. Product Selection

Five commercially available desk–chair systems (Brands A–E) were purposively selected to represent diverse ergonomic configurations, reclining mechanisms, material compositions, and sensory characteristics common in Chinese primary schools (Figure 2). All products conformed to GB/T 28007-2011 safety standards and were marketed for combined study–napping purposes [12].



Figure 2. Representative nap-compatible Desk-Chair Products (Brands A–E).

2.2.2. Participants

A total of 122 participants were recruited as follows:

- The 112 primary school students (grades 4–6, aged 10–12) represented the primary end-users;
- Ten educational ergonomics experts, each with over five years of experience in school furniture evaluation, child ergonomics, or product certification.

The study specifically targeted students in grades 4–6, as this group represents the primary users of nap-compatible classroom chairs in Chinese schools. Through preliminary field research, we observed that nap-compatible chairs are rarely adopted in grades 1–3,

where some schools provide nap beds or have shorter schedules that reduce the need for in-class naps. Conversely, many middle-school students reside in dormitories and can return to their rooms for midday rest. Therefore, the 4–6 grade cohort (40 fourth-graders, 42 fifth-graders, and 30 sixth-graders) was selected as the most representative user group. In addition, ten ergonomics experts were included to ensure evaluation stability and professional validity, although the main emphasis of the study remains on students' experiential feedback.

Students were stratified by grade and gender, then rotated across chair samples in a Latin square design to mitigate order effects. Written informed consent was obtained from school administrators and guardians, and the study followed the British Educational Research Association guidelines [15].

2.2.3. Scenario Configuration

The following three task scenarios simulated daily classroom activities:

- Scenario 1—Writing (8 min): Upright seated tasks (reading/writing), assessing postural support and desk usability;
- Scenario 2—Adjustment (2 min): Transition to nap-compatible state, assessing ease and intuitiveness of transformation;
- Scenario 3—Napping (10 min): Reclined rest with eyes closed, evaluating material comfort, tactile perception, and emotional resonance.

Environmental conditions (lighting, temperature, noise) were held constant. The rotation ensured that each participant experienced all products in all scenarios (Figure 3).

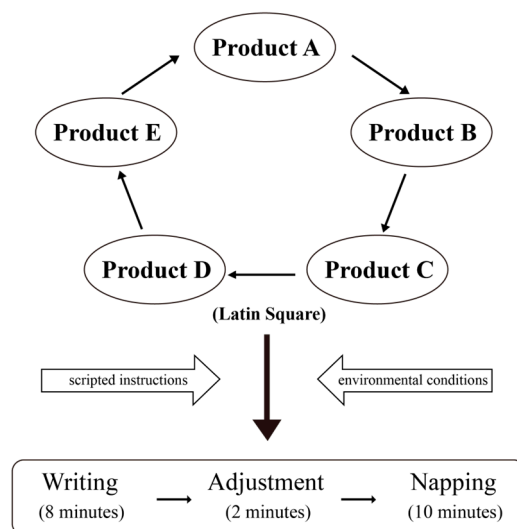


Figure 3. Experimental procedure for student evaluations, including scenario sequence and product rotation logic.

2.2.4. Evaluation Procedure

After each scenario, participants rated the chair across all eight PLEASURC dimensions using a 10-point Likert scale. Experts, observing from outside, scored the products using predefined criteria and behavioral checklists aligned with each dimension. Each participant completed 15 evaluations (5 products × 3 scenarios), producing a rich dataset for conflict-sensitive weighting, tensor decomposition, and machine learning analysis.

2.3. Conflict-Sensitive Adaptive Weighting Method

To objectively integrate expert and student evaluations, a conflict-sensitive dynamic weighting method was adopted. This method introduced perceptual divergence into the

weighting process, allowing adaptive adjustment based on intergroup agreement at each evaluation dimension [16,17].

In this study, conflict-sensitive dynamic weighting refers to a divergence-aware integration mechanism that balances the relative influence of experts and students depending on their level of agreement. When student and expert scores align closely, both sides contribute equally; when perceptual conflict is high, expert input is given more weight to ensure stability and compliance with professional standards. In this way, the weighting is not static but adapts dynamically to the extent of disagreement.

Let e_i and s_i denote the score distributions of experts and students for dimension i . The Jensen–Shannon Divergence (JSD) [18], a symmetric and bounded metric derived from entropy theory, is used to quantify their perceptual discrepancy:

$$m_i = \frac{1}{2}(e_i + s_i) \quad (1)$$

$$JSD_i = \frac{1}{2}D_{KL}(e_i||m_i) + \frac{1}{2}D_{KL}(s_i||m_i) \quad (2)$$

where D_{KL} denotes the Kullback–Leibler divergence.

To convert the divergence metric into a weighting coefficient, a *Sigmoid* function is applied for nonlinear mapping:

$$\alpha_i = \frac{1}{1 + \exp(-k \times JSD_i)} \quad (3)$$

Here, $\alpha_i \in (0, 1)$ represents the weight assigned to expert input for dimension i , and K is a hyperparameter controlling the sensitivity of the function. We conducted sensitivity analysis across a broader range ($K = 5\text{--}20$) and found that setting $K = 10$ yielded stable convergence across scenarios and balanced responsiveness to expert–student divergence, while avoiding overweighting when perceptual differences were minor.

The final fused score for dimension i is given by:

$$w_i = \alpha_i \times w_i^{\text{expert}} + (1 - \alpha_i)w_i^{\text{student}} \quad (4)$$

The method dynamically reflects rating consistency between experts and students. Greater consistency increases fusion reliability, while divergence shifts weight toward expert input. This enhances the professionalism, stability, and adaptability of the composite weights.

Additionally, JSD values serve as perceptual conflict indicators and can be incorporated as auxiliary variables in scenario sensitivity analysis and interaction modeling.

2.4. Statistical Analysis

Data processing was performed using Python 3.11 (pandas, tensorly, shap), SPSS 26.0, and Microsoft Excel 365. The analysis included the following:

- Scale reliability: Cronbach’s α (≥ 0.70) and item–total correlations to validate the PLEASURC scale.
- Descriptive analysis: Mean scores for eight PLEASURC dimensions—Ergonomics, Safety, Learnability, Usability, Perception, Aesthetics, Relatability, and Comfort—were computed to profile product performance.
- Scenario sensitivity analysis: A third-order tensor (Dimension $\times$ Product $\times$ Scenario) was constructed from user ratings across writing, adjustment, and napping scenarios. PARAFAC extracted latent factors to identify sensitivity patterns and product–scenario alignment.

- Perceptual interaction modeling: An XGBoost regression model was trained using Comfort as the target variable. SHAP-enhanced interpretation was used to analyze nonlinear interactions among perceptual and functional dimensions, revealing key predictors of comfort perception.

2.5. Integrated Analytical Pipeline

First, conflict-sensitive weighting (Jensen–Shannon divergence) adaptively integrates expert and student evaluations, ensuring that both perspectives are represented in proportion to their agreement level. Second, tensor decomposition (PARAFAC) is applied to the three-way dataset (Dimension $\times$ Product $\times$ Scenario), revealing scenario-specific sensitivity patterns—e.g., writing tasks emphasize Learnability and Usability, whereas napping emphasizes Comfort and Aesthetics. Third, explainable machine learning (XGBoost + SHAP) is used to quantify the contribution of each dimension to Comfort, which is modeled as the target variable. The model was trained with five-fold cross-validation (train/test split $\approx$ 8:2), achieving $R^2 = 0.832$ (training) and $R^2 = 0.711$ (testing), indicating good predictive accuracy and generalizability. In plain terms: the weighting mechanism decides whose opinion counts more, tensor decomposition shows when different factors matter most, and SHAP analysis explains what specifically drives comfort and why. These three methods are complementary, yielding results that are interpretable, replicable, and practically actionable.

2.6. Qualitative Classification Board

In addition to quantitative scoring, a complementary classification board was employed to provide binary pass/fail judgments (work vs. not work) on essential safety and usability items. This qualitative process ensured that no chair could be rated highly overall if it failed basic requirements. The checklist covered five key aspects: (1) reliable locking during posture transformation, (2) absence of finger entrapment risks, (3) rounded and child-safe edges, (4) ability of students to independently restore the chair to the learning position, and (5) stability under weight shift. These criteria were scored as pass/fail by experts and recorded alongside quantitative ratings. In practice, the classification board acted as a safety veto layer: any failure was marked as “not work,” regardless of other scores.

3. Results

3.1. Reliability and Validity of the PLEASURC Scale

The qualitative classification board confirmed that all five products passed the essential safety and usability requirements (locking, entrapment, edges, restoration, stability). No “not work” outcomes were observed, indicating that all tested chairs met the minimum functional and safety thresholds before quantitative scoring.

A total of 122 complete response sets were collected, covering three distinct scenarios—writing, adjustment, and napping—across five nap-compatible desk-chair products. Each participant provided ratings on the eight PLEASURC dimensions from both student and expert perspectives.

Internal consistency of the PLEASURC scale was validated using Cronbach’s alpha ($\alpha = 0.750$), indicating satisfactory overall reliability. Item–total correlation coefficients ranged from 0.155 for Learnability to 0.737 for Safety [19]. Higher correlations were observed for Ergonomics ($r = 0.577$), Perception ($r = 0.585$), and Safety ($r = 0.737$), signifying stronger alignment with the overall user evaluation. In contrast, weaker correlations for Learnability ($r = 0.155$) and Aesthetics ($r = 0.319$) point to greater perceptual independence within those dimensions.

This result affirms the discriminant validity of the PLEASURC dimensions and supports their inclusion as distinct but complementary components of a holistic ergonomic evaluation framework. The moderate inter-item correlations also reflect the model’s sensitivity to multi-domain perceptions—especially relevant for multifunctional school furniture.

3.2. Scenario-Specific Sensitivity Patterns

To investigate variations in user sensitivity across different usage scenarios, three-dimensional rating data were represented as a tensor $\chi \in \mathbb{R}^{S \times P \times D}$, and analyzed using the PARAFAC (Parallel Factor Analysis) model [20,21]. The decomposition is expressed as:

$$\chi_{spd} \approx \sum_{r=1}^R a_{sr} \cdot b_{pr} \cdot c_{dr} \quad (5)$$

where χ_{spd} denotes the average user rating for product p under scenario s on dimension d , and $A = [a_{sr}]$, $B = [b_{pr}]$, and $C = [c_{dr}]$ are the factor matrices corresponding to scenarios, products, and dimensions, respectively.

Given that the writing and napping scenarios represent distinct use states—task-focused and rest-oriented, respectively—their latent factors were selected for targeted comparison. In this study, the tensor rank was set to $R = 2$ after testing alternative ranks ($R = 2-4$). Model fit plateaued beyond rank 2, while higher ranks reduced interpretability; thus, $R = 2$ was chosen as the optimal balance of accuracy and parsimony. Conceptually, the two extracted factors can be intuitively viewed as a “task fluency axis” and a “rest comfort axis,” reflecting the dual nature of classroom chair use.

As shown in Figure 4, the writing scenario was primarily associated with high loadings in Learnability (0.58), Usability (0.54), Perception (0.38), and Ergonomics (0.35). In this context, a higher loading value means that a given dimension contributes more strongly to shaping user perceptions in that scenario. Thus, the writing state is driven mainly by operational fluency and posture support, reflecting a user focus on operational fluency and posture support during academic tasks.

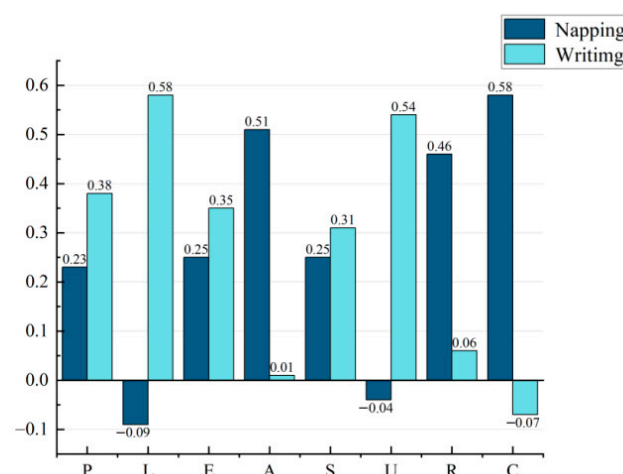


Figure 4. Scene sensitivity map.

In contrast, the napping scenario emphasized Comfort (0.58), Aesthetics (0.51), and Relatability (0.46), underscoring the importance of sensory and emotional dimensions in rest-oriented use.

These scenario-specific patterns suggest that dual-function classroom chairs should be ergonomically optimized for both task execution and restorative comfort, depending on contextual demands.

Brand-level scenario compatibility was derived from dot-product analysis (Figure 5), showing that Brand B performed best under writing conditions, while Brand A was preferred for napping. This insight provides data-informed guidance for product optimization based on usage context.

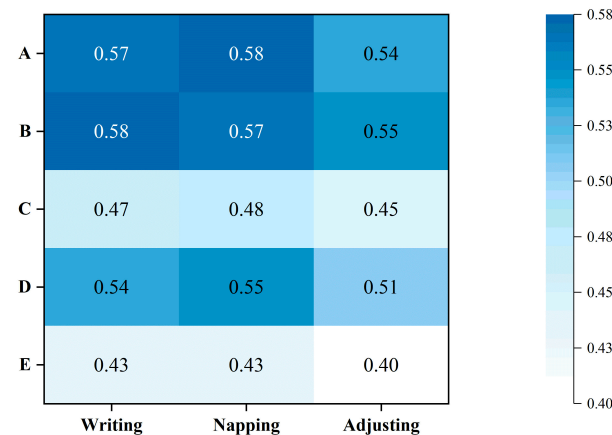


Figure 5. Scene adaptation matrix.

3.3. Perceptual Interactions and Predictors of Comfort

SHAP (SHapley Additive Explanations) was applied to interpret the XGBoost regression model, which used Comfort as the target variable, reflecting overall user experience within the PLEASURC framework. SHAP, based on Shapley values, provides consistent local and global attribution for nonlinear models and is well-suited for explaining complex perceptual systems [22].

The XGBoost model was optimized via recursive feature selection and evaluated through five-fold cross-validation, yielding strong predictive performance with $R^2 = 0.832$ on the training set and $R^2 = 0.711$ on the test set [23,24].

To capture nonlinear integration of multisensory and functional perceptions, we extended the eight PLEASURC dimensions with three derived interaction terms:

$$\text{Material-Ergonomic Interaction} = P \times E$$

$$\text{Safety-Function Interaction} = S \times U$$

$$\text{Aesthetics-Emotional Interaction} = A \times R$$

where P, E, S, U, A, and R represent Perception, Ergonomics, Safety, Usability, Aesthetics, and Relatability, respectively. These interaction terms were not new dimensions of the framework, but analytical constructs introduced solely within the SHAP model to examine whether cross-domain synergies provide additional predictive value for Comfort. Thus, while the framework remains grounded in eight dimensions grouped into four domains, the predictive model effectively operated with eleven explanatory variables (the eight original dimensions plus the three interaction terms).

Quantitatively, the top four predictors together contributed 68.39% of the total model attribution for Comfort. Within this set, the Material–Ergonomic interaction alone explained 24.72%, Safety contributed 22.55%, Aesthetics 17.22%, and Perception 3.90%. These results indicate that sensory and affective dimensions (Aesthetics and Perception) are not secondary but comparable in weight to traditional structural considerations (Safety and Ergonomics).

As shown in Figure 6, the Material–Ergonomic Interaction emerged as the most influential factor in shaping comfort perceptions, followed by Safety, Aesthetics, and

Perception. Figure 7 further illustrates that high values of these dimensions consistently contributed to positive comfort ratings.

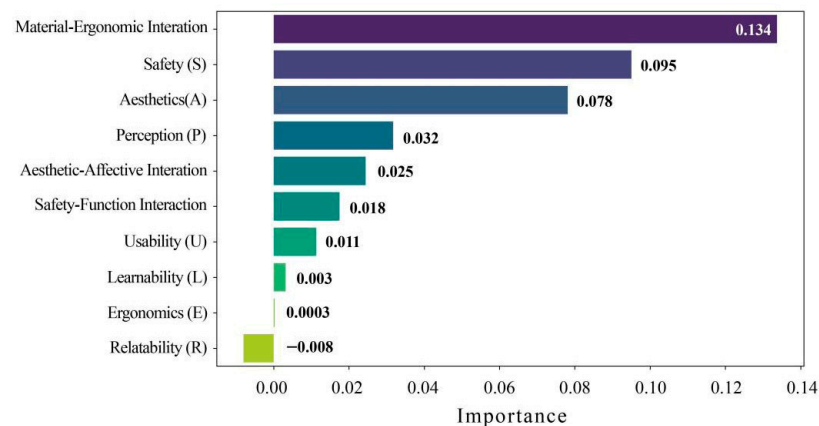


Figure 6. SHAP global feature importance ranking chart.

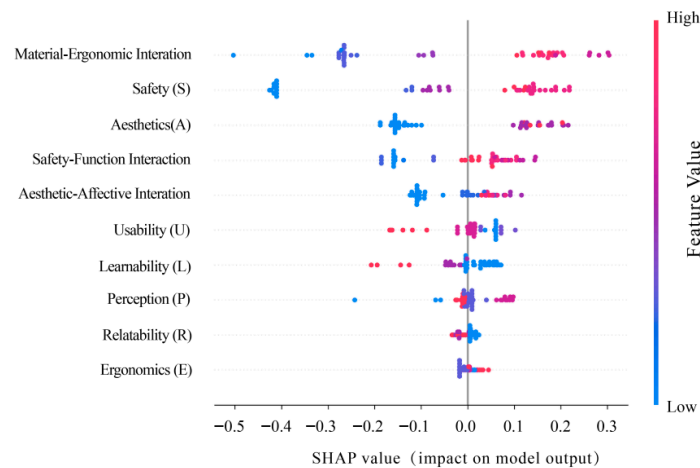


Figure 7. Effect of dimensional characteristics on Comfort (C).

These findings highlight the synergistic effects between structural support, tactile experience, and emotional design—key considerations in improving nap-compatible school furniture.

3.4. Product Performance Comparison

3.4.1. Scenario-Specific Brand Performance Analysis

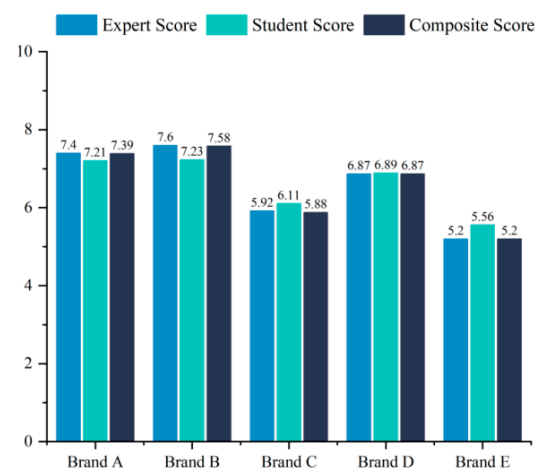
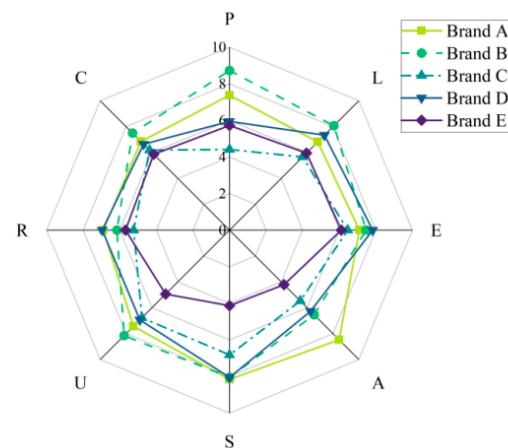
To evaluate the performance of each brand across representative classroom scenarios, a dynamic weighting model was applied to compute expert, student, and composite scores under napping, writing, and adjustment scenarios (Table 2). Figures 8 and 9 visualize performance profiles across the eight PLEASURC dimensions, providing a multi-perspective comparison.

Among the five tested brands, clear performance differences emerged (Table 2). Brand B achieved the highest overall performance (composite score: 7.58/10), particularly excelling in writing and adjustment scenarios—indicating strong functional and operational attributes. Brand A stood out in the napping scenario, reflecting superior emotional resonance and sensory compatibility.

By contrast, Brand D demonstrated consistent, moderate performance across all contexts, suggesting balanced but unremarkable ergonomic attributes. Brands C and E exhibited lower scores across most dimensions and scenarios, likely due to structural limitations, weaker sensory appeal, or poor usability.

Table 2. Weighted composite scores across eight dimensions for five brands.

Brand	Scenario	P	L	E	A	S	U	R	C	Expert Score	Student Score	Composite Score
Brand A	Writing	5.42	6.82	5.24	9.57	8.16	7.43	6.87	4.85	6.83	6.88	6.80
	Adjustment	7.37	7.98	7.10	6.61	8.16	8.37	6.89	6.85	7.37	6.92	7.42
	Napping	5.42	6.82	5.23	9.58	6.31	7.43	4.90	4.85	6.33	6.58	6.32
	Composite	7.37	6.82	7.10	8.47	8.16	7.43	6.90	6.85	7.40	7.21	7.39
Brand B	Writing	6.81	8.07	5.66	6.78	8.04	8.15	6.15	5.50	6.90	6.99	6.89
	Adjustment	8.72	9.17	7.47	5.02	8.04	9.17	6.34	7.50	7.70	6.82	7.68
	Napping	6.84	8.07	5.64	6.65	6.37	8.15	4.34	5.50	6.43	6.60	6.44
	Composite	8.72	8.07	7.47	6.54	8.04	8.15	6.16	7.50	7.60	7.23	7.58
Brand C	Writing	3.42	5.65	5.04	5.31	6.82	6.77	5.24	4.37	5.38	5.77	5.33
	Adjustment	4.41	6.78	6.46	4.41	6.82	7.58	6.07	6.18	6.10	5.80	6.09
	Napping	3.42	5.65	5.03	5.28	5.29	6.77	4.07	4.37	5.00	5.48	4.99
	Composite	4.41	5.65	6.46	5.47	6.82	6.77	5.24	6.18	5.93	6.11	5.88
Brand D	Writing	4.62	7.34	6.03	6.62	8.05	6.93	5.96	4.64	6.30	6.63	6.27
	Adjustment	5.94	8.37	7.83	4.62	8.05	7.98	6.36	6.64	6.98	6.57	6.97
	Napping	4.62	7.34	6.08	6.76	6.38	6.93	4.36	4.64	5.85	6.26	5.89
	Composite	5.94	7.34	7.83	6.27	8.05	6.93	5.96	6.64	6.87	6.89	6.87
Brand E	Writing	4.61	5.95	5.01	4.30	4.13	4.94	5.69	3.72	4.78	5.19	4.79
	Adjustment	5.72	6.98	6.11	3.22	4.13	4.94	6.00	4.86	5.25	5.20	5.24
	Napping	4.61	5.95	5.01	4.30	3.24	4.94	4.00	3.72	4.45	4.93	4.47
	Composite	5.72	5.95	6.11	4.21	4.13	4.94	5.69	4.86	5.20	5.56	5.20

**Figure 8.** Comparison of expert, student, and total scores for five products.**Figure 9.** Radar plot of overall PLEASURC scores for Brands A–E.

Taken together, these results highlight that Brand B is the most versatile across task-oriented contexts, while Brand A is the most competitive for rest-oriented use, offering complementary strengths depending on design priorities.

These findings highlight that scenario-specific ergonomic requirements vary significantly, and that no single design excels universally. As such, differentiated design optimization and market positioning strategies are recommended for manufacturers seeking to improve product relevance and contextual fit.

3.4.2. Perceptual Divergence Between Experts and Students

Perceptual divergence between experts and students was quantitatively assessed using expert weight coefficients (α) derived from the Jensen–Shannon divergence–based fusion model. These coefficients indicate the relative influence of expert evaluations for each dimension: higher α values (approaching 1) reflect expert dominance and greater intergroup disagreement, while lower values (closer to 0) suggest student-driven perceptions and higher consensus.

Figure 10 presents perceptual conflict heatmaps based on Jensen–Shannon divergence between student and expert ratings across the eight PLEASURC dimensions and three use scenarios. Numerically, higher divergence values indicate stronger disagreement between the two groups, while lower values reflect closer alignment. Visually, darker or more intense colors on the heatmap correspond to greater perceptual conflict. In practical terms, the heatmap can be intuitively read as a “tension map,” where shaded areas mark dimensions that require reconciliation in design and evaluation.

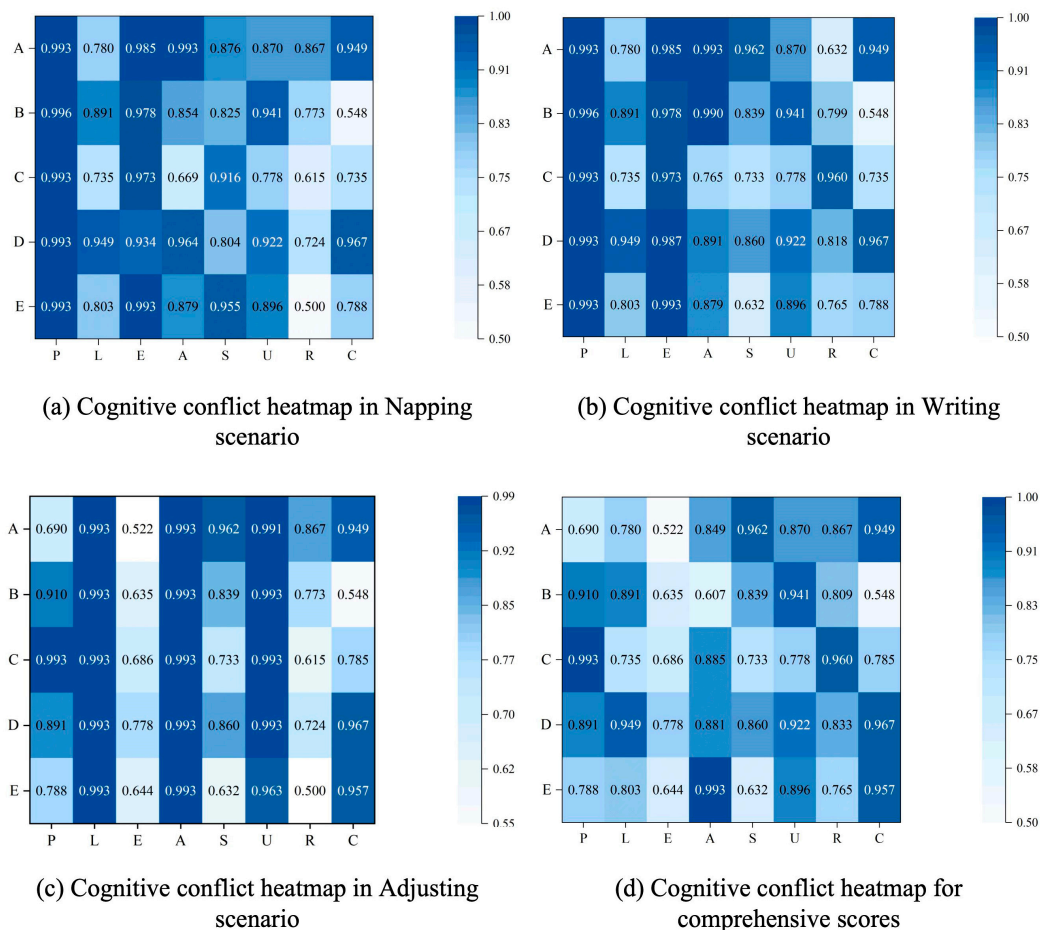


Figure 10. Expert–student perceptual conflict and expert-weight map (α). (a) Cognitive conflict heatmap in Napping scenario; (b) Cognitive conflict heatmap in Writing scenario; (c) Cognitive conflict heatmap in Adjusting scenario; (d) Cognitive conflict heatmap for comprehensive scores.

In the napping scenario, α values were generally high, indicating strong expert influence (Figure 10a). Perception and Ergonomics showed consistently high α values across

brands, highlighting expert dominance in evaluating structural support and sensory feedback. Relatability had the lowest α values, particularly for Brands C (0.615) and E (0.500), suggesting greater subjectivity among users. Notable brand-specific divergence was also observed in Safety and Aesthetics—for example, Brand C had a relatively low α of 0.669 in Aesthetics, indicating student-driven impressions. Overall, divergence was most evident in functional and sensory dimensions.

In the writing scenario, expert dominance remained pronounced (Figure 10b). Perception and Ergonomics again exhibited the highest α values, confirming the expert-led nature of ergonomic assessment. In contrast, Comfort showed lower divergence, with α values of 0.548 (Brand B) and 0.735 (Brand C), suggesting a stronger influence from student perception. Divergence in Learnability and Safety across brands indicated greater user involvement in interpreting functional clarity and security. These findings emphasize the need to balance expert attention to operational logic with user expectations for intuitive interaction.

In the adjustment scenario, α values were more dispersed, indicating greater context dependency (Figure 10c). Learnability, Aesthetics, and Usability all had α values exceeding 0.96, reflecting high expert consensus. However, Ergonomics showed lower α values for Brands A (0.522) and B (0.635), suggesting divergent user perceptions of postural adaptability. The scenario elicited expert focus on structural and functional attributes, while students prioritized interactive comfort and emotional experience.

In the comprehensive evaluation across all scenarios (Figure 10d), α values exhibited a balanced distribution. Brand A showed strong expert influence in Aesthetics ($\alpha = 0.849$) and Safety ($\alpha = 0.962$), reflecting heightened sensitivity to visual and safety attributes. Brand B displayed minimal divergence in Comfort ($\alpha = 0.548$) and moderate consensus in Relatability ($\alpha = 0.809$), indicating broad acceptance. Brand C showed the largest expert–student discrepancy in Perception ($\alpha = 0.993$) and Relatability ($\alpha = 0.960$), suggesting potential perceptual bias. Brand D maintained consistent α values across dimensions, particularly in Learnability and Usability (both $\alpha > 0.90$), indicating expert-driven assessments of interaction features. Brand E exhibited the lowest α in Safety ($\alpha = 0.632$), with student perceptions dominating.

Collectively, the dimensions showing the greatest perceptual divergence were Relatability, Learnability, and Aesthetics, underscoring the importance of these factors in child-centered ergonomic design. These findings suggest that while experts dominate in structural and functional domains, students play a decisive role in evaluating emotional, intuitive, and aesthetics qualities. The brand-wise divergence profiles offer actionable insights for tailoring design strategies to balance professional standards with authentic user experience.

3.5. Feature Integration from Data Insights

To validate the framework’s design utility, we applied the insights from Sections 3.2–3.4 to redesign the top-performing product (Brand B), focusing on Relatability, Learnability, Aesthetics, and Material–Ergonomic synergy.

3.5.1. Design Improvements Informed by Data Insights

Statistical and SHAP-enhanced analyses revealed that Relatability, Learnability, and Aesthetics—along with interaction terms such as Material–Ergonomic synergy and Safety feedback—were the most influential dimensions driving comfort perception. A new prototype, was therefore developed with focused improvements in the following aspects:

- Segmented lumbar support and a curved ergonomic backrest to enhance perceived safety and postural adaptation;

- Four-dimensional adjustable seat and leg rest, supporting intuitive operation and physical adaptability;
- Memory foam-based pressure relief zones and skin-friendly tactile materials for enhanced sensory comfort;
- Visually unified color scheme and curved styling, optimized for aesthetics coherence and emotional affinity (Figure 11).



Figure 11. Redesigned product informed by data-driven ergonomic indicators.

3.5.2. User Feedback and Comparative Results

To evaluate the effectiveness of the redesigned model, a follow-up mini-study was conducted with 3 experts and 20 students (aged 10–12), who experienced the redesigned product under the same three scenarios as the original study. PLEASURC scores were collected and compared against those of Brand B (Table 3).

Table 3. A comparison of user ratings (mean scores) between Brand B and the redesigned product on the most affected dimensions.

Dimension	Brand B	Redesigned Product	Δ Improvement
Relatability (R)	6.16	7.28	+1.12
Learnability (L)	8.07	8.65	+0.58
Aesthetics (A)	6.54	7.62	+1.08
Comfort (C)	7.60	8.12	+0.52
Safety (S)	8.04	8.48	+0.44

Participants reported that the redesigned chair was easier to operate, more visually appealing, and better aligned with their resting needs, particularly in the napping scenario. These results provide empirical confirmation that the theoretical model can effectively inform tangible design improvements in real-world classroom furniture.

4. Discussion

4.1. Validation and Theoretical Implications of the PLEASURC Model

The combined application of conflict-sensitive weighting, scenario-specific tensor decomposition, and SHAP-enhanced interpretation provides robust empirical validation for the proposed PLEASURC framework. Reliability analysis (Cronbach's $\alpha = 0.750$), systematic expert–student divergence patterns, and predictive accuracy ($R^2 > 0.70$) confirm its structural soundness and sensitivity to perceptual differences [25].

Unlike conventional task-oriented ergonomic frameworks, PLEASURC integrates structural, operational, perceptual, and affective domains, offering a multidimensional,

child-centered evaluation structure tailored to dual-purpose classroom environments [26]. The redesign experiment further demonstrated external validity, with targeted improvements yielding measurable gains in Relatability (+1.12), Aesthetics (+1.08), Comfort, and Safety [27,28].

Finally, the inclusion of a binary classification board enhances qualitative robustness by ensuring that essential safety and usability standards are not overlooked. These converging lines of evidence establish PLEASURC as both a reliable evaluation tool and a practical guide for product optimization.

4.2. Expert–Student Divergence and Design Implications

The Jensen–Shannon divergence-based adaptive weighting method effectively quantified cognitive conflicts between expert-driven assessment criteria and student experiential feedback. Dimensions such as Relatability and Learnability exhibited lower α -values, indicating a stronger influence of student perceptions, whereas Usability and Ergonomics were more expert-dominated [29].

This divergence highlights a critical design challenge: reconciling technical performance and safety compliance with users' intuitive and emotional experiences. For example, while experts may rate Safety highly based on structural compliance, students may feel less secure if tactile feedback or operational cues are lacking. These results underscore the need for participatory design strategies and iterative user feedback loops in school furniture development—particularly for features affecting daily comfort and trust [30].

4.3. Key Design Weaknesses and Optimization Priorities

Results from tensor decomposition and SHAP-enhanced analysis revealed scenario-specific mismatches and design inefficiencies across brands. For instance, Brand B performed best in task-oriented contexts (writing and adjustment), while Brand A achieved higher scores in rest-oriented conditions, reflecting differentiated ergonomic strengths. At the feature level, Material–Ergonomic synergy and Safety perception emerged as dominant predictors of comfort, exposing limitations in physical feedback and sensory design among lower-performing brands [31].

Common weaknesses included the following:

- Insufficient edge rounding and feedback mechanisms, reducing dynamic safety assurance;
- Complex or unintuitive transformation procedures, undermining usability and learnability;
- Rigid plastic surfaces lacking thermal responsiveness, impairing tactile satisfaction.

To address these issues, optimization should focus on:

- Multimodal safety feedback (visual, tactile, auditory cues);
- Cognitively efficient transformation pathways, informed by human–computer interaction models (e.g., GOMS) [32];
- Adaptive, sensory-friendly materials (e.g., memory foam, thermoregulating textiles) that enhance comfort while maintaining structural support [33].

The redesigned prototype, informed by these priorities, demonstrated significant improvements in user evaluations, validating the transferability of model-driven insights into practical design solutions [34].

4.4. Practical Applications, Policy Implications and Scope of Application

Before discussing broader applications, it is important to clarify the scope of this study. The evaluation was conducted on five commercially available nap-compatible desk–chair systems in Chinese primary schools, with 122 participants (112 students in grades 4–6 and 10 educational ergonomics experts). The scenarios focused on short-term tasks of writing, adjustment, and napping, under controlled environmental conditions. Therefore, while the

framework is replicable and transferable—its eight dimensions, weighting method, and analysis pipeline can be applied to other furniture types and user cohorts—generalization beyond this demographic and context should be made cautiously. Broader adoption would require site-specific data collection and, ideally, longitudinal validation in different cultural or educational settings.

With these boundaries clarified, the practical applications of the PLEASURC framework become evident. It offers a replicable, evidence-based evaluation tool for nap-compatible classroom furniture, enabling integration of expert insights with authentic student feedback. Its eight-dimensional structure supports data-driven procurement, design benchmarking, and quality assurance for manufacturers, school administrators, and policymakers [35].

In China, the implementation of midday rest policies has accelerated demand for dual-function desk–chair systems that can accommodate both learning and napping in space-constrained classrooms [36]. The PLEASURC model provides a standardized evaluation reference that aligns product development with these policy goals.

Internationally, although multifunctional rest-integrated classroom furniture is less common, the framework is adaptable to boarding schools, after-school programs, and special education environments, especially where spatial efficiency and student well-being intersect [37]. For regulators, PLEASURC could serve as the foundation for ergonomic certification systems and child-centered procurement standards, fostering healthier and more inclusive educational spaces [38].

4.5. Limitations and Future Research Directions

Several limitations of the present study warrant consideration. First, participants were restricted to grades 4–6 (ages 10–12), as lower grades often use nap beds or have shorter schedules, and middle school students typically rest in dormitories. Although this group represents the main users of nap-compatible classroom chairs, future studies should expand to broader age cohorts. Second, evaluations were short-term, we plan to conduct daily and semester-long tracking of the optimized prototype (Section 3.5) to assess durability and evolving perceptions. Third, the study was limited to Chinese classrooms, and cross-cultural differences in rest policies, classroom layouts, and user expectations remain unexplored; these contextual factors should be addressed in future validation.

To address these limitations and advance the applicability of the PLEASURC framework, future research should consider the following directions:

- Age-and gender-specific modeling: Incorporate developmental variability and inclusive anthropometric data to tailor furniture ergonomics across broader student populations;
- Real-time behavioral and physiological tracking: Integrate tools such as posture sensors, electromyography (EMG), or video-based coding to assess continuous interaction patterns, fatigue buildup, and micro-behavioral cues;
- Cross-cultural and contextual validation: Evaluate the PLEASURC model in diverse educational and cultural settings to enhance its global relevance and transferability, particularly in under-resourced or non-Western environments;
- AI-enhanced adaptive learning spaces: Explore the integration of the PLEASURC framework with AI-driven smart classroom systems, enabling dynamic adjustment of furniture features (e.g., recline angles, support surfaces) in response to students' physical state or usage behavior;
- Longitudinal and outcome-based validation: Conduct extended studies that link ergonomic evaluations to learning outcomes, attentional regulation, or well-being metrics, to build stronger evidence for educational policy and procurement.

Among these proposed directions, cross-cultural validation and semester-scale longitudinal studies appear most immediately feasible, as they build directly on the present design. By contrast, AI-driven adaptive classroom systems remain a longer-term vision requiring infrastructure development.

5. Conclusions

This study presents a conflict-sensitive, multidimensional evaluation (PLEASURC) framework for nap-compatible classroom chairs and examines its performance in a primary-school setting. By combining expert–student adaptive weighting, tensor decomposition, and SHAP-interpreted predictive modeling, the approach provided interpretable evidence on scenario-specific sensitivities and feature contributions to Comfort. In cross-validated analyses, the model achieved good predictive accuracy, and a framework-guided redesign was associated with improvements in several user-relevant dimensions. These findings suggest that the method can inform evidence-based evaluation and product iteration where midday rest is part of routine school practice.

The results reveal that perceptual divergence—particularly in Relatability, Learnability, and Aesthetics—plays a decisive role in shaping comfort perceptions, often surpassing structural or safety factors. Moreover, interaction effects, such as Material–Ergonomic synergy and Safety feedback, emerged as key predictors of overall user experience across different classroom scenarios. These insights underscore the importance of balancing structural robustness with emotional resonance and sensory appeal in school furniture design.

The PLEASURC framework advances educational ergonomics by bridging structural, operational, perceptual, and affective domains, offering a more holistic understanding of student–furniture interaction. Beyond academic research, its applicability extends to the following:

- Product design iteration, enabling targeted improvements based on data-driven insights;
- Evidence-based procurement, assisting schools and policymakers in selecting ergonomically and emotionally optimized products;
- Health-oriented educational policy development, supporting the creation of sustainable and student-centered learning environments.

Importantly, our results also indicate that improvements in comfort, usability, and emotional acceptance are tightly linked to durability, correct use, and life-cycle efficiency. By turning multidimensional evidence into actionable design and procurement criteria, the proposed framework provides a practical route toward healthy, resource-efficient, and policy-aligned classroom environments.

Looking ahead, future work should pursue longitudinal and cross-cultural validation, incorporate physiological and behavioral tracking, and explore integration with AI-powered adaptive classroom systems. These developments will enhance the personalization and responsiveness of classroom furniture, contributing to broader goals in student well-being, learning efficiency, and sustainable educational infrastructure.

By offering a methodologically rigorous and practically relevant evaluation paradigm, this research provides not only a theoretical foundation for ergonomics in educational contexts but also a direct pathway toward innovation in school furniture design—ensuring that classroom environments can truly support both learning and rest in a balanced, sustainable manner.

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